



- $$U_n = \left(\frac{n+1}{2n}\right)^2. \quad [4]$$

[illegible]

- (b) Use a similar method to find, in terms of n , a lower bound L_n for $\int_0^1 x^3 dx$. [4]

This image shows a blank sheet of white paper with horizontal dashed lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no other markings, text, or illustrations on the page.

- (c) Find the least value of n such that $U_n - L_n < 10^{-3}$. [2]

[illegible]